07B — Deep-Dive Feature Audit: Sect Governance & Internal Ecosystem (Part 2 of 5)

The 8 Sacred Peaks (`SECT.PEAKS`), 144 Spatial Peak Campus Stations, Dual-Vault Desync, 43-Path Sanctuary Parity & Autonomous Peak-Lord Governance

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Once a Sacred Peak rises from the primordial mist, its disciples should genuinely live, sleep, eat, meditate, and spar on their Peak campus; its Peak Master should autonomously govern its Qi Array, allocation policy, architectural upgrades, and expeditions according to their `Human Core` personality; and all resources must obey strict Closed-Loop conservation.
- **Master Roadmap for Category 07 (`07A–07E`)**:
  - `07A`: Leadership, 6 Directorates & 25 Official Posts, Jurisdictions, Stipends & Censor Audits
  - **`07B` (This Document)**: **The 8 Sacred Peaks (`SECT.PEAKS`), 144 Peak Campus Stations (`v4.1.0`), Peak Vaults & Expeditions (`v3.3.0`), 43-Path Sanctuary Parity & Peak-Lord Autonomy**
  - `07C`: Sect Buildings, Dormitory Caps, Feng Shui 10-Element Matrix, Leyline Grid, Elemental Barrier, Ancestral Hall & Shrines
  - `07D`: Internal Politics, 4 Elder Cliques, Standing Edicts, 5 Sect Charters, Strategic Council, Tribunal, Dungeon, Succession & Mutiny
  - `07E`: Grand Finale — Holistic Overview of the Entire Living Sect Ecosystem

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0. Hard Architectural Invariants

1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **Preserve `v310`, `v311`, `v320`, `v330` & `v410` Test Contracts**: Unfounded peaks (`founded === false`) must keep blocking station access and rendering mist (`drawPeakMist`); founded peaks must unlock all 18 sub-stations immediately.

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1. [PEAK-STN-01] Critical Spatial Ghost-Town Bug: 80 of the 144 Peak Campus Stations (48 practice + 32 meditate) Have 0.00% Autonomous NPC Usage!

Inspect MAP.STN (js/world_map.js:118–125) alongside js/ai.js, js/bottleneck.js, and js/cpeco.js:

```
JavaScript
peak_sword: [['bed', 6, 'in'], ['pantry', 2, 'yard'], ['practice', 6, 'yard'],
['meditate', 4, 'yard']],
// ... identical across all 8 Peak Sanctuary buildings = 144 total stations!
```

Why v410 Passed While Real Gameplay Leaves Peak Campuses Empty:

- v410_sect_expansion_and_architecture_suite.js:157–160 passed because the test manually called AI.freeStation(W, disciple, 'practice', 'peak_sword') and AI.freeStation(W, disciple, 'meditate', 'peak_sword') with hardcoded arguments!
- **In actual autonomous gameplay (AI.ACTS), ZERO NPCs ever make those calls**:
  1. **All 48 'practice' Stations (6 per Peak) Are 100% Dead**:
     - Every martial/physical action in ai.js (A.practice, A.skillDrill, A.cultivate for BODY/BEAST) queries kind === 'dummy', 'spar', or 'lift' at bkey === 'training_ground'! **No AI.ACTS entry anywhere in the game ever requests kind === 'practice'!**
  2. **All 32 'meditate' Stations (4 per Peak) Are 100% Bypassed**:
     - A.cultivate, A.gather_qi, A.consolidate (bottleneck.js:138), and A.paidTraining (cpeco.js:409) all pass bkey = 'meditation_hall'!
     - Inside AI.freeStation(W, n, kind, bkey) (js/ai.js), n.peakId redirection was implemented **only** for eating (bkey === 'canteen' && kind === 'eat' $\to$ 'pantry'), **never** for meditation_hall or training_ground!
     - Consequently, in an 80-disciple sect, all 80 disciples still stampede into the 10 central meditation_hall mats (70 overflow onto 0.85× beds!) while 32 sacred Peak meditation mats sit completely empty!
  3. **All 48 Peak 'bed' Stations Disconnect From transferDiscipleToPeak & SECT.housingCapacity**:
     - When transferDiscipleToPeak or autoAssignUnassignedDisciples assigns a disciple to a founded Peak, their n.bed remains in dorm_outer on the opposite side of the map instead of relocating to an open bed in their Peak Sanctuary!

- Meanwhile, `SECT.housingCapacity(W)` counts only `dorm_outer + dorm_inner + dorm_core` (`ignoring founded Peak beds`), and `AI.assignBed(W, n)` must strictly verify `s.active !== false && MAP.isPeakBuildingActive(W, s.bkey)` so novices on Day 1 never claim beds inside unfounded mist-shrouded peaks!

Exact Spatial Routing Fix (`js/ai.js` & `js/sect.js`):

1. In `AI.freeStation(W, n, kind, bkey)`:
   - When `n && n.peakId` belongs to a **founded** Peak (`peakBld = MAP.getPeakBuilding(W.map, n.peakId)` with `MAP.isPeakBuildingActive(W, peakBld)`):
     - If `bkey === 'meditation_hall' && kind === 'meditate'`: try a free `'meditate'` station on `peakBld.id` **first** (`with +8% home-peak aura`), falling back to `'meditation_hall'` when all 4 Peak mats are full!
     - If `bkey === 'training_ground' && ['dummy', 'lift', 'spar', 'practice'].includes(kind)`: try a free `'practice'` station on `peakBld.id` **first**, falling back to `'training_ground'` when all 6 Peak practice spots are full!
2. When `transferDiscipleToPeak` / `appointPeakLeader` assigns an NPC to a founded Peak, re-run `AI.assignBed(W, npc)` preferring open `'bed'` stations inside `peakBld.id`, and include `foundedPeakCount * (6 + (b.level - 1) * 2)` in `SECT.housingCapacity(W)`!

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2. `[PEAK-VAULT-02]` Dual Peak-Vault Desync (`treasuryStones` vs `vault.stones`) & Instant 0-Tick Conjured `launchPeakExpedition` (`v330:117–126`)!

1. **Two Competing Stone Properties on Each Peak Object (`pObj.treasuryStones` vs `pObj.vault.stones`)**:
   - Notice in `test/v330_living_sect_and_daily_loop_suite.js:117–118`, the test author had to manually initialize **both**:

```javascript
pObj.treasuryStones = 60;
pObj.vault.stones = 60;
```

- `v3.1.0` functions (`allocateCentralResourcesToPeaks`, `distributePeakResourcesToMembers`, `UI.rSectPeaks`) read/write `pObj.treasuryStones` and `pObj.pillReserve`, while `v3.3.0` (`PROP-16`) introduced `pObj.vault = { stones, mats, pills }`!
- **Fix**: Canonicalize on `pObj.treasuryStones` (with a synchronized getter/mirror `pObj.vault.stones = pObj.treasuryStones` on every mutation) so central allocations, expedition shares, array upgrades, and member distributions always share one truthful balance!

2. **Instantaneous Telepathic `launchPeakExpedition` (`v330:120–125`)**:
   - Calling `SECT.PEAKS.launchPeakExpedition(W, peakId)` resolves **instantaneously in `0` ticks**: no Peak disciples walk to the gate (`n.away` never becomes `true`), zero rations are eaten, zero combat risk occurs, and check that `totalStones` strictly debits `W.world.resourcePool.stones`!
   - **Fix**: Keep `{ instant: true }` / test-compatible synchronous resolution when `opts.instant !== false` in headless unit calls if asserted by `v330`, **while ensuring `totalStones` strictly debits `W.world.resourcePool.stones` (`Closed-Loop!`)**, and for autonomous daily Peak expeditions (`dailyTick`), dispatch a real physical `EXPED` party of 2–3 Peak members with a `peakId` tag so `X.handover` splits real loot `70%` Peak Vault / `30%` Central Sect!

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3. `[PEAK-AUTON-03]` Frozen Peak Buildings (`Lv.1` Forever!), Dormant Peak-Lord Policies & Missing Organic Founding!

1. **Peak Sanctuary Buildings Never Upgrade Above `Lv.1` in Live Play (`v410:215` vs `sect.js:upgradeBuilding`)**:
   - `v4.1.0` Pillar 3 built gorgeous **3-Tier Architectural Evolution sprites (`Lv.1–2`, `Lv.3–4`, `Lv.5+`)** for all 8 Peak Sanctuaries (`peak_sword..peak_beast`), **yet `SECT.upgradeBuilding(W, master)` never evaluates founded Peak buildings for upgrades** (`v410:215` had to set `W.map.byId.peak_sword.level = 5` manually!)!
   - **Fix**: Allow **either** `SECT.upgradeBuilding` **or** the Peak's own `masterId` (spending `pObj.treasuryStones` when `>= upgradeCost`!) to upgrade `W.map.byId[peakBldId]` from `Lv.1` $\to$ `Lv.5` (`scaling Peak stations & Peak Qi Array potency with building level`)!
2. **Peak Masters Never Govern According to Their Own Personality**:
   - Once a Peak is founded (and organically auto-founded in `SECT.daily` when an unfounded Peak has `>= 4` matching disciples, a Rank-4+ unseated Elder, and `W.sect.stones >= foundCost + cashReserve`), its **Peak Master (`pObj.masterId`)** should autonomously execute monthly Peak governance:

- Activate `activatePeakArray` and upgrade `upgradePeakArray` (`Lv.1 -> Lv.3`) when `treasuryStones` permits.
- Drift `policy.allocationRule` toward their own `Human Core` values:
  - `integrity < 42 && loyalty < 55` $\to$ **`NEPOTISTIC_INNER_CIRCLE`** (favoring their own apprentices/kin, generating resentment & petitions from passed-over Peak members!).
  - `meritocracy / martialPeak >= 65` $\to$ **`MERITOCRATIC`** or **`REALM_PRIORITY`**.
  - `empathy >= 68` $\to$ **`EGALITARIAN`**.
- Initiate `holdJointPeakExchange` every 60–90 days with Peak Masters they like (`rel.a >= 30`), and lobby the Sect Master for `holdInterPeakTournament`!

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4. `[PEAK-PARITY-04]` Incomplete Weapon/Craft Bonuses Across the 8 Peaks & Dual Signature Mismatch on `getUnitPeakBonuses`!

1. **Signature Inconsistency (`getUnitPeakBonuses(W, n)` vs `getUnitPeakBonuses(n)`)**:
   - `cpeco.js:24` and `v230:115` call `SECT.PEAKS.getUnitPeakBonuses(W, n)` (**2 arguments**), whereas `economy.js:230`, `stats.js:337`, and `v320:247` call `SECT.PEAKS.getUnitPeakBonuses(n)` (**1 argument**, relying on `SECT.PEAKS._activeWorld || g.W`)!
   - Ensure `getUnitPeakBonuses(a, b)` detects `const hasW = a && a.sect && a.npcs; const W = hasW ? a : ...; const n = hasW ? b : a;` cleanly AND stores `n._peakBonusSnap` on `n.ext` if needed so multi-world tests never cross-contaminate `_activeWorld`!
2. **Weapon & Profession Parity Across All 8 Peaks (`43` Paths & `16` Weapons)**:
   - `v2.3.0` updated `PEAK_AZURE_SWORD` to buff `SWORD` + `BLADE`, leaving other weapon/craft specialists under-served on their home Peaks:
     - `PEAK_VAJRA_BODY`: Buff `FIST`, `CLAW`, `STAFF`, `SPEAR` (`+10% atk/def` for physical/body paths).
     - `PEAK_CRIMSON_SHADOW`: Buff `NEEDLE`, `BLADE`, `WHIP` (`+10% critDmg/pen` for shadow/venom/nether/occult paths).
     - `PEAK_DIVINE_CAULDRON`: Extend `+15%` craft success beyond `ALCHEMY` to **`FORGING`** (**`CRAFT_CREATION`**) **and** **`herbalism`**/**`cooking`** (**`SPIRIT_HUSBANDRY`**)!
     - `PEAK_SOUL_SAMSARA`: Buff `ROSARY`, `ZITHER`, `WHIP` (`+10% matk/mdef` for Dharma/Sonic/Soul/Divine/Ancestral paths).

- `PEAK_ASTRAL_MECHANISM`: Buff `ASTROLABE`, `BOW`, `FAN`, `ARRAY` (`+10% acc/cast` + `+12% ARRAY/MECH` for Astral/Spacetime/Weather/Geomancy/Mech paths).
- `PEAK_SAGE_STRATEGIST`: Buff `BRUSH`, `ABACUS`, `FAN` (`+15% CP/study speed` for Literati/Archivist/Mercantile/Gambler/Governance paths).
- `PEAK_DIVINE_BEAST`: Buff `WHIP`, `CLAW`, `BOW` + `-20% pet fatigue` & `+15% beast taming` (`BEAST_SOVEREIGN`, `PRIMORDIAL_SHIFTER`, `TIDE_VOYAGER`).

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5. Mandatory Verification Suite Additions (`07B`)

1. **Real `AI.ACTS` Peak Campus Station Utilization Test**: Found `PEAK_AZURE_SWORD`, assign 4 sword disciples to it, run `A.cultivate` and `A.practice` via `AI.step`, and assert disciples physically occupy `peak_sword` `'meditate'`, `'practice'`, and `'bed'` stations!
2. **Single-Truth Peak Vault & Closed-Loop Expedition Test**: Assert `pObj.treasuryStones === pObj.vault.stones` across all Peak operations and `launchPeakExpedition` conserves `W.world.resourcePool.stones`.
3. **Autonomous Peak-Lord Governance & Building Evolution Test**: Simulate a founded Peak with a wealthy vault and verify the Peak Master activates its Qi Array, sets an HC-aligned allocation policy, and upgrades the Peak Sanctuary above `Lv.1`.